

1. Introduction

We will produce:

- Background
 - Motivation
 - Research gap
 - Problem statement
 - Contributions
 - Paper organization
-

2. Related Work

Instead of reproducing the thesis literature review, we will synthesize it into themes such as:

- Hallucination mitigation
- Knowledge graph reasoning
- Graph-constrained decoding
- Prompt-based reasoning
- Research gap summary

including a comparison table showing why DCA-Trie differs from GCR, DoG, and related methods.

3. Proposed Framework

This will become the core of the paper.

We will include:

- Overall architecture
- Mathematical formulation
- TypeOracle
- DCA-Trie
- Semantic Irrelevance Ratio (SIR)
- Algorithms
- Complexity analysis

(For publication, strengthen it by discussing:

- worst-case complexity,
- expected complexity,
- memory complexity,
- pruning efficiency.

If DCA-Trie reduces the search space, quantify that reduction.

)

4. Theoretical Analysis

This section will strengthen the scientific contribution.

For example:

- Correctness of TypeOracle
- Constraint validity
- Search-space reduction
- Why semantic pruning preserves structural faithfulness
- Why improved constraint quality does not necessarily imply improved reasoning accuracy

This section is largely absent from the dissertation and will significantly improve the paper.

5. Experimental Design

We will clearly describe:

- Datasets
- Baselines
- Evaluation metrics
- Experimental settings
- Research questions

without presenting results.

6. Experimental Results

This section will present only the evidence.

For example:

- Overall performance
- Comparison with GCR and DoG
- Runtime
- Memory
- Search-space reduction
- SIR evaluation

Each table and figure will have concise interpretation.

7. Extended Analysis

Following the style of papers like APKGC, we can include:

- Ablation study
- Oracle analysis
- Sensitivity analysis
- Efficiency analysis
- Robustness evaluation

This is where we demonstrate *why* the framework behaves as observed.

8. Discussion

This is where I think we can make your paper much stronger than the dissertation.

Rather than saying:

"Our method achieved X."

we will discuss:

- Why DCA-Trie reduces semantic irrelevance.
- Why reducing irrelevant paths did not increase Hits@1.
- What this reveals about graph-constrained decoding.
- Practical implications for future LLM reasoning systems.
- Limitations of the framework.

This section should leave reviewers with a scientific insight, not just performance numbers.

9. Conclusion

A concise conclusion covering:

- Main findings
- Contributions
- Limitations
- Future work

Section	Pages	What should be included
Title, Abstract & Keywords	1	Strong title, 200–250 word abstract, 5–7 keywords.
1. Introduction	3	Background, motivation, problem statement, research gap, contributions, paper organization.
2. Related Work	3–4	Critical comparison of hallucination mitigation, KG reasoning, constrained decoding, prompt-based methods, and a comparison table highlighting the gap.
3. Proposed Framework	5–6	Overall architecture, mathematical formulation, TypeOracle, DCA-Trie, SIR, algorithms, computational complexity.
4. Theoretical Analysis	2–3	Correctness analysis, structural faithfulness, search-space reduction, theoretical discussion of oracle behavior.
5. Experimental Design	2	Datasets, baselines, implementation details, evaluation metrics, research questions, hardware/software configuration.
6. Experimental Results	3–4	Performance comparison, efficiency, runtime, memory, quantitative evaluation.
7. Extended Analysis	2–3	Ablation study, oracle analysis, robustness, sensitivity analysis, error analysis.
8. Discussion	2–3	Scientific interpretation, implications, limitations, why the results occurred, comparison with the broader literature.
9. Conclusion	1	Summary, contributions, limitations, future work.
References	3–5	Around 60–100 references, depending on the breadth of the review.

2. Related Work (≈ 3–4 pages)

The literature review should **not** be organized like a textbook.

Instead organize around competing approaches.

2.1 Hallucination Mitigation

Brief review of

- RAG
- CoT
- Self-RAG

Focus on their limitations.

2.2 Knowledge Graph Reasoning

Discuss

- KGQA
 - Multi-hop reasoning
 - Graph reasoning
-

2.3 Constrained Decoding

Review

- GCD
- Outlines
- GCR
- DoG

This section should naturally lead to
none adapts constraints according to semantic context.

2.4 Research Gap Summary

One figure or table.

Example

Method	Structural Constraint	Dynami c	Semanti c	Oracle
GCR	✓	×	×	Static
DoG	✓	Partial	×	Dynamic Topology
RouterKGQ A	×	✓	✓	Retrieval
Proposed	✓	✓	✓	Context Adaptive

This table alone will convince reviewers you understand the literature.